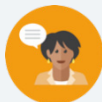


7.H3 Area of a Triangle Using the Sine Ratio

Lesson Introduction

 Benchmark	MA.912.T.1.4 Solve mathematical problems involving finding the area of a triangle given two sides and the included angle.		 Learning Target	I can find the area of a triangle using the sine ratio.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	How can I find the area of a triangle using the sine ratio?			
 Lesson Narrative	In this lesson, students work to solve mathematical problems of finding the area of a triangle, given two sides and the included angle, using the sine ratio. Students will begin by exploring the area of a triangle using trigonometry. They will then formally learn the three different formulas for finding the area of a triangle using sine.			
 Component Summary	7.H3.1 Warm-Up (~5 min.)	Observations of mathematics everywhere (<i>SE p. 440</i>)	7.H3.3 Guided Practice (10-15 min.)	Finding the area of a triangle using trigonometry (<i>SE p. 442</i>)
	7.H3.2 Exploration (15 min.)	Deriving the formulas for using the sine ratio to find the area of a triangle (<i>SE p. 440</i>)	7.H3.4 Your Turn! (10-15 min.)	Finding the area of a triangle using trigonometry (<i>SE p. 442</i>)
	7.H3.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 443</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Calculators	
	Additional Preparation This is an Honors lesson. Only students enrolled in Geometry Honors should be required to complete this lesson.			

 Teacher Tips	Common Misconceptions - Students may mistakenly invert the values in the sine ratio. - Students may not use the order of operations correctly when evaluating the formulas for finding the area of a triangle using sine.	Things to Consider The purpose of 7.H3.2 Exploration is for students to connect what they know about finding the area of a triangle, given all the necessary side information, to situations where two sides and the included angle are given. Try not to give too much away during the Exploration, so students can make the appropriate connections.
	Literacy and Language Routines Take Turns This lesson does not have a Guided Instruction. Consider leveraging the “Take Turns” instructional routine during the 7.H3.3 Guided Practice. This way, students can take turns in the work of the activity, being able to explain the rationale for their choices, and listen to another’s rationale to deepen their understanding.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure During this lesson, students will learn to connect what they have learned about finding the area of a triangle to finding the area of a triangle using trigonometry. Students will relate those previously learned concepts to new concepts while looking for similarities among problems.
	 Differentiate Instruction Consider adding in a flashcard or rapid-fire activity for students. Create pictures of triangles where the area could be solved using a basic area formula and triangles where students would need to use trigonometry. This can help students quickly recognize when one method would be required versus another. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

7.H3.1 Warm-Up: History of Mathematics

Component Overview: Students will review information about Hannah Fry, an English mathematician, and respond to two related questions.



7.H3.1 Warm-Up: History of Mathematics

Hannah Fry is an English mathematician who works alongside physicists, mathematicians, computer scientists, architects, and geographers to study the patterns in human behavior. Fry's research applies to a wide range of social problems and questions, from shopping and transport to urban crime, riots, and terrorism.

Fry has also written two popular math books: *The Mathematics of Love: Patterns, Proofs, and the Search for the Ultimate Equation* and *The Indisputable Existence of Santa Claus*.

Fry is quoted as saying, "There is a mathematical angle to almost anything."

1. Interpret this quote in your own words.

Answers will vary. You can find a way make almost anything into a story problem.

2. What is a specific real-world example where mathematics is present?

Answers will vary. Finding the area of a room to carpet it.



Consider having students do additional reading and/or research on Hannah Fry as a way to enhance this activity.

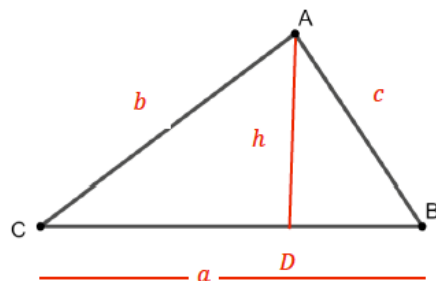
7.H3.2 Exploration: Area of a Triangle

Component Overview: Students will explore the area of a triangle. Students will connect situations where they can solve using a basic area formula to where the use of trigonometry would be required.



7.H3.2 Exploration: Area of a Triangle

1. $\triangle ABC$ is shown.



a. Use $\triangle ABC$ to do the following.

- Draw an altitude from point A to side $\overline{CB}$
- Label the altitude as h .
- Label the point where the altitude intersects side $\overline{BC}$ as D.
- Label the sides as $AB = c$, $BC = a$, and $AC = b$.

The altitude creates two right triangles inside $\triangle ABC$.

b. Write the equation that can be used to find the area of $\triangle ABC$.

$$A = \frac{1}{2}ah$$

c. Complete each equation using right triangle trigonometry.

$\triangle ADB$	$\triangle ADC$
$\sin(B) = \frac{h}{c}$	$\sin(C) = \frac{h}{b}$



The purpose of the Exploration activity is for students to connect what they know about finding the area of triangle, given all the necessary side information, to situations where two sides and the included angle are given. Try not to give too much away during the Exploration, so students can make the appropriate connections.



- How did you determine the ratio for question 1c?
- How can you verify that you are correct?

7.H3.2 Exploration: Area of a Triangle (continued)

- d. Rewrite each equation in terms of h .

$\triangle ADB$	$\triangle ADC$
$h = c \sin B$	$h = b \sin C$

- e. Substitute each expression into the formula,
 $A = \frac{1}{2}ah$.

$\triangle ADB$	$\triangle ADC$
$A = \frac{1}{2}ac \sin B$	$A = \frac{1}{2}ab \sin C$

- f. What do you notice about the two equations?
Answers will vary. Both equations have $\frac{1}{2}a$.

2. $\triangle ABC$ is shown.

- a. Find the area of $\triangle ABC$ given that $m\angle C = 41^\circ$, $AC = 55$, and $BC = 73$.

$$A = \frac{1}{2}(55)(73) \sin 41; A \approx 1317 \text{ square units}$$

- b. Find the area of the same triangle given that $m\angle A = 90^\circ$, $AC = 55$, and $AB = 48$.

$$A = \frac{1}{2}(48)(55) = 1320 \text{ square units}$$

- c. Discuss with your partner why the formula $A = \frac{1}{2}bh$ could also be used when given $m\angle A = 90^\circ$, $AC = 55$, and $AB = 48$. Summarize your discussion.

The formula $A = \frac{1}{2}bh$ can be used because this is a right triangle. The two measures given were the legs which represent the base and the height.



Consider jotting down what you hear for question 1f when circulating during this activity. Use that as a starting point for a discussion at the conclusion of this activity.



Have students discuss their answers with multiple partners to come to a finalized consensus among different perspectives and approaches.

7.H3.3 Guided Practice: Area of a Triangle

Component Overview: Students will complete two practice questions with finding the area of a triangle using sine.



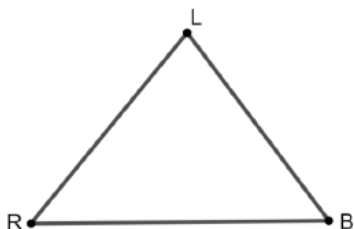
7.H3.3 Guided Practice: Area of a Triangle



The formulas for finding the area of a triangle using sine are:

$$A = \frac{1}{2}bc \sin A \quad A = \frac{1}{2}ac \sin B \quad A = \frac{1}{2}ab \sin C$$

1. $\triangle RLB$ is shown. $RL = 14$, $RB = 16.5$, and $m\angle LRB = 50^\circ$. Find the area of the triangle.



$$A = \frac{1}{2}(16.5)(14) \sin 50; \quad A = 88.478 \text{ square units}$$

2. In $\triangle FPB$, $FP = 9$, $PB = 18$, and $m\angle FPB = 117^\circ$. Find the area of the triangle.

$$A = \frac{1}{2}(9)(18) \sin 117; \quad A = 72.172 \text{ square units}$$



Consider leveraging the “Take Turns” instructional routine during the Guided Practice. This way, students can take turns in the work of the activity, being able to explain the rationale for their choices, and listen to another’s rationale to deepen their understanding.



- How can you determine the best method for finding the area in question 2?
- Why would trigonometry be required to find the area?

7.H3.4 Your Turn!

Component Overview: Students will complete two practice questions with finding the area of a triangle using trigonometry.



7.H3.4 Your Turn!

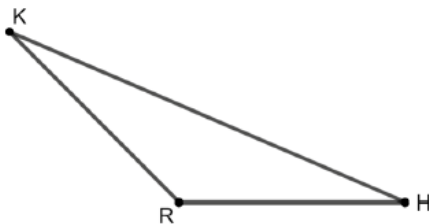
1. In $\triangle NBR$, $NB = 20$, $NR = 18.6$, and $m\angle RNB = 43^\circ$. Find the area of the triangle.

$$A = \frac{1}{2}(20)(18.6) \sin 43; \quad A = 126.852 \text{ square units}$$



Consider using this time to pull together a small group of students who may have been struggling during the Exploration and/or the Guided Practice of this lesson to provide additional support with these questions in a small group setting.

2. $\triangle KRH$ is shown. $KR = 18$, $RH = 16$, and $m\angle KRH = 135^\circ$. Find the area of the triangle.



$$A = \frac{1}{2}(18)(16) \sin 135; \quad A = 101.823 \text{ square units}$$



What is the relationship between finding the area of a triangle with knowing the side values, and finding the area of a triangle using trigonometry?

7.H3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



7.H3.5 Wrap-Up: Cool Down

1. What are the similarities and differences between finding the area of a triangle and finding the area of a triangle using the sine ratio?

Answers will vary.



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.

ISBN 978-1-952882-74-6



ISBN 978-1-955898-75-1



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